Space Probe in the Earth-Moon system

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Newton's Gravity and 2nd law $f = ma$ imply a probe at $\mathbf{p}(t) = (x(t), y(t))$ in rotating coordinates with the earth and moon fixed at $\mathbf{p}_e = (-\mu, 0)$ and $\mathbf{p}_m = (1-\mu, 0)$ satisfies

$$\boxed{\begin{aligned} \ddot{x} - 2\dot{y} &= x - (1-\mu)\frac{x+\mu}{d_e^3} - \mu\frac{x-1+\mu}{d_m^3} \\ \ddot{y} + 2\dot{x} &= y - (1-\mu)\frac{y}{d_e^3} - \mu\frac{y}{d_m^3}. \end{aligned}}$$

where the moon-earth mass ratio $\mu = \frac{m_m}{m_e} = \frac{1}{82.45}$, $d_e = ||\mathbf{p}(t) - \mathbf{p}_e||$, and $d_m = ||\mathbf{p}(t) - \mathbf{p}_m||$.

Rotations

Measuring distance and time in convenient units the rotation matrix $R(t)$ and skew matrix Ω

$$R(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix} \quad \text{and} \quad \Omega = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

(satisfying $\ddot{R}(t) = -R(t)$ and $\dot{R}(t) = \Omega R(t)$) give clean representations of probe position $\mathbf{r}(t)$ and the moon and earths circular orbits about the center of mass at the origin

$$\mathbf{r}(t) = R(t)\mathbf{p}(t) \quad \mathbf{r}_m(t) = R(t)\mathbf{p}_m \quad \text{and} \quad \mathbf{r}_e(t) = R(t)\mathbf{p}_e.$$

Differentiating twice gives

$$\ddot{\mathbf{r}} = R\ddot{\mathbf{p}} + 2\dot{R}\dot{\mathbf{p}} + \ddot{R}\mathbf{p} = R\big(\ddot{\mathbf{p}} + 2\Omega\dot{\mathbf{p}} - \mathbf{p}\big)$$

Note: Distance in LD (Earth-Moon 342,402 km) and time in month$/2\pi \approx 4.5$ days.

Newton's Laws

If the probe has mass m Newton's law is

$$m\ddot{\mathbf{r}} = -\frac{Gmm_e}{d_e^3}(\mathbf{r} - \mathbf{r}_e) - \frac{Gmm_m}{d_m^3}(\mathbf{r} - \mathbf{r}_m)$$

where $d_e = ||r - r_e|| = ||p - p_e||$ and $d_m = ||r - r_m|| = ||p - p_m||$. In our special units $Gm_m m_e = 1$ so canceling m gives

$$\ddot{\mathbf{r}} = -\frac{(1-\mu)}{d_e^3}(\mathbf{r} - \mathbf{r}_e) - \frac{\mu}{d_m^3}(\mathbf{r} - \mathbf{r}_m)$$

Since $\mathbf{r} = R\mathbf{p}$, $\mathbf{r}_e = R\mathbf{p}_e$, $\mathbf{r}_m = R\mathbf{p}_m$ and $\ddot{r} = R\left(\ddot{\mathbf{p}} + 2\Omega\dot{\mathbf{p}} - \mathbf{p}\right)$

$$R\left(\ddot{\mathbf{p}} + 2\Omega\dot{\mathbf{p}} - \mathbf{p}\right) = -\frac{(1-\mu)}{d_e^3}R(\mathbf{r} - \mathbf{r}_e) - \frac{\mu}{d_m^3}R(\mathbf{r} - \mathbf{r}_m)$$

Simplification

Since R^{-1} exists the definitions of Ω, p, p_m, and p_e give our ODEs

$$\begin{pmatrix} \ddot{x} - 2\dot{y} - x \\ \ddot{y} + 2\dot{x} - y \end{pmatrix} = -\frac{1-\mu}{d_e^3} \begin{pmatrix} (x - 1 + \mu) \\ y \end{pmatrix} - \frac{\mu}{d_m^3} \begin{pmatrix} (x + \mu) \\ y \end{pmatrix}$$

Note: The *Falcon Heavy* launch mass $m_f \approx 1.4 \times 10^6$ kg satisfies $m_f / m_m < 10^{-16}$ and $m_f / m_e < 10^{-18}$. Current spacecraft have no measurable effect on earth or moon orbits.

`https://orbital-mechanics.space/the-n-body-problem/circular-restricted-three-body-problem.html`